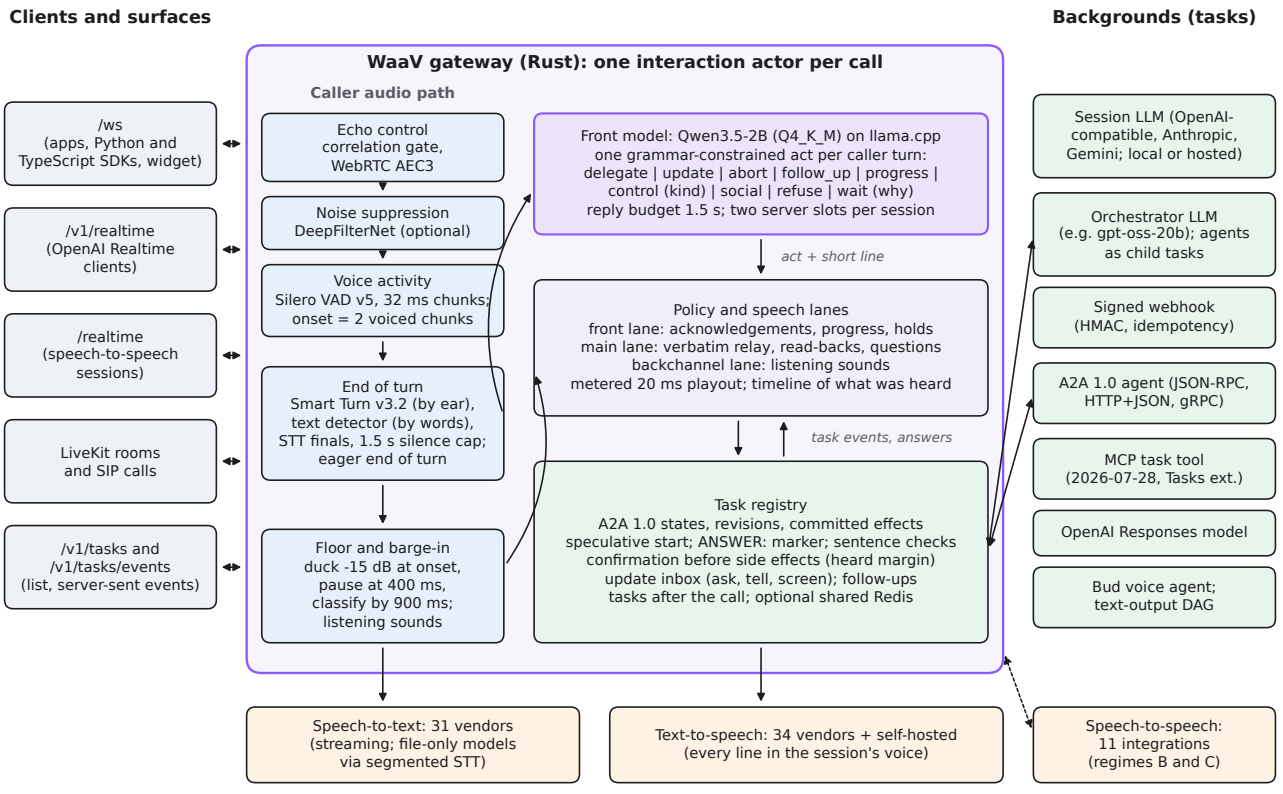
Clients and surfaces
Backgrounds (tasks)
WaaV gateway (Rust): one interaction actor per call
Caller audio path
/ws
(apps, Python and
TypeScript SDKs, widget)
/v1/realtime
(OpenAI Realtime
clients)
/realtime
(speech-to-speech
sessions)
LiveKit rooms
and SIP calls
/v1/tasks and
/v1/tasks/events
(list, server-sent events)
Echo control
correlation gate,
WebRTC AEC3
Noise suppression
DeepFilterNet (optional)
Voice activity
Silero VAD v5, 32 ms chunks;
onset = 2 voiced chunks
End of turn
Smart Turn v3.2 (by ear),
text detector (by words),
STT finals, 1.5 s silence cap;
eager end of turn
Floor and barge-in
duck -15 dB at onset,
pause at 400 ms,
classify by 900 ms;
listening sounds
Front model: Qwen3.5-2B (Q4_K_M) on llama.cpp
one grammar-constrained act per caller turn:
delegate | update | abort | follow_up | progress |
control (kind) | social | refuse | wait (why)
reply budget 1.5 s; two server slots per session
act + short line
Policy and speech lanes
front lane: acknowledgements, progress, holds
main lane: verbatim relay, read-backs, questions
backchannel lane: listening sounds
metered 20 ms playout; timeline of what was heard
task events, answers
Task registry
A2A 1.0 states, revisions, committed effects
speculative start; ANSWER: marker; sentence checks
confirmation before side effects (heard margin)
update inbox (ask, tell, screen); follow-ups
tasks after the call; optional shared Redis
Session LLM (OpenAI-
compatible, Anthropic,
Gemini; local or hosted)
Orchestrator LLM
(e.g. gpt-oss-20b); agents
as child tasks
Signed webhook
(HMAC, idempotency)
A2A 1.0 agent (JSON-RPC,
HTTP+JSON, gRPC)
MCP task tool
(2026-07-28, Tasks ext.)
OpenAI Responses model
Bud voice agent;
text-output DAG
Speech-to-text: 31 vendors
(streaming; file-only models
via segmented STT)
Text-to-speech: 34 vendors + self-hosted
(every line in the session's voice)
Speech-to-speech:
11 integrations
(regimes B and C)